

# Collaborative Cache-Aided Relaying Networks

## Reproducible Experiment Results

Paper: Collaborative cache-aided relaying networks:

Performance evaluation and system optimization

Authors: Shunpu Tang, Ke He, Lunyuan Chen, Lisheng Fan,  
Xianfu Lei, and Rose Qingyang Hu

IEEE Journal on Selected Areas in Communications, 41(3):706-719, 2023

Generated from the Python reproduction scripts in this repository.



Fig. 1: Outage probability versus relay transmit power.

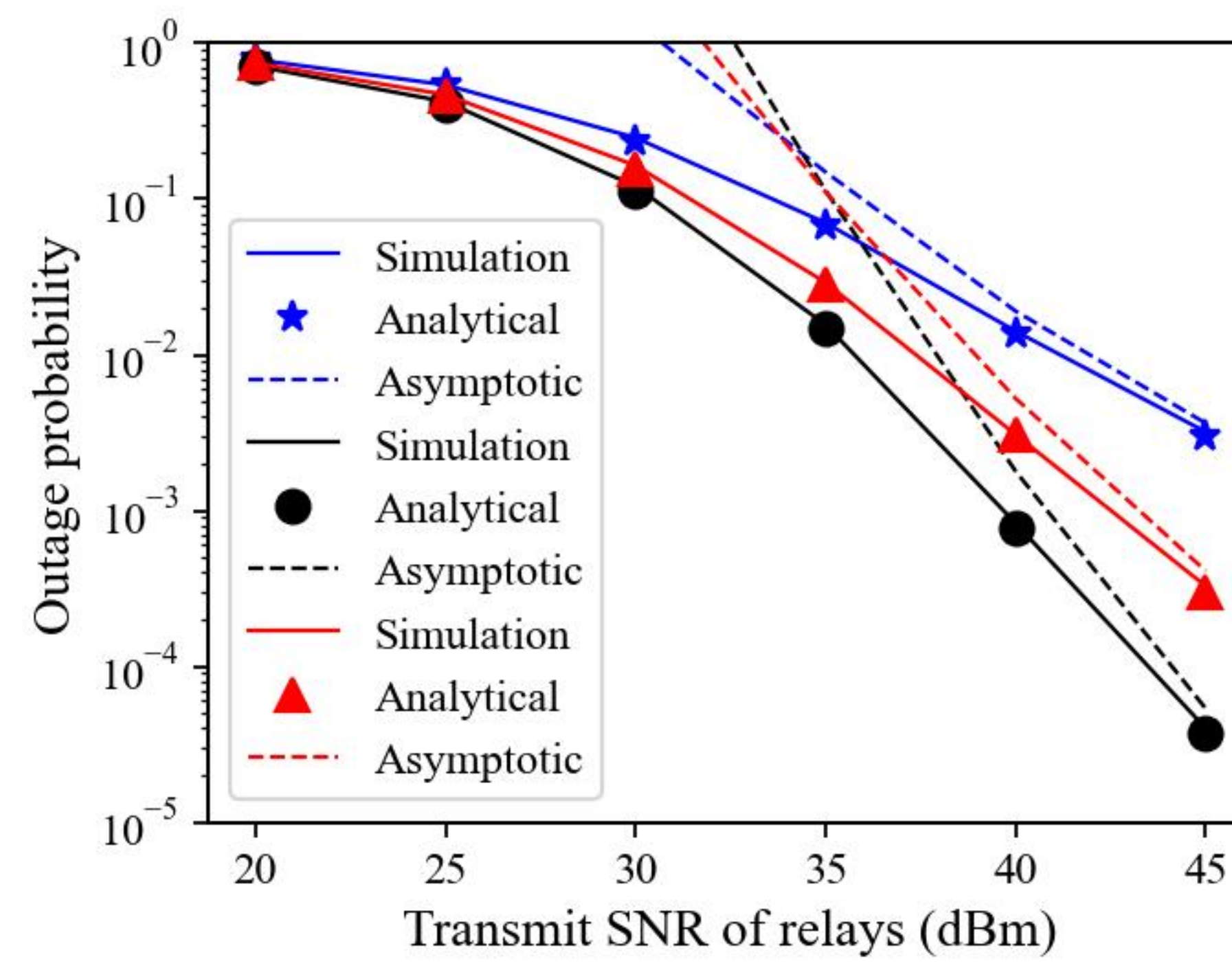




Fig. 2: Outage probability versus BS transmit power.

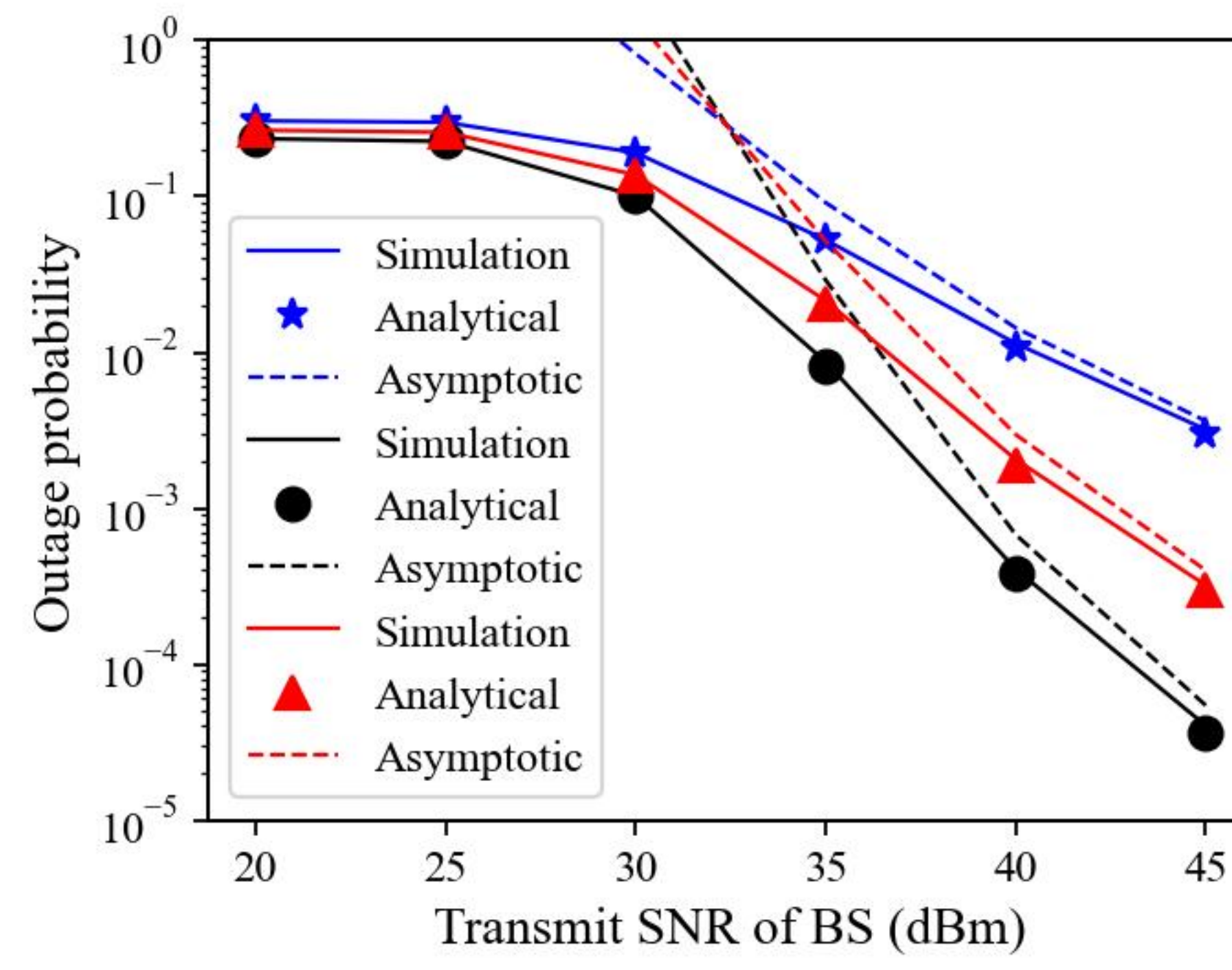




Fig. 3: Soft-BCD convergence behavior.

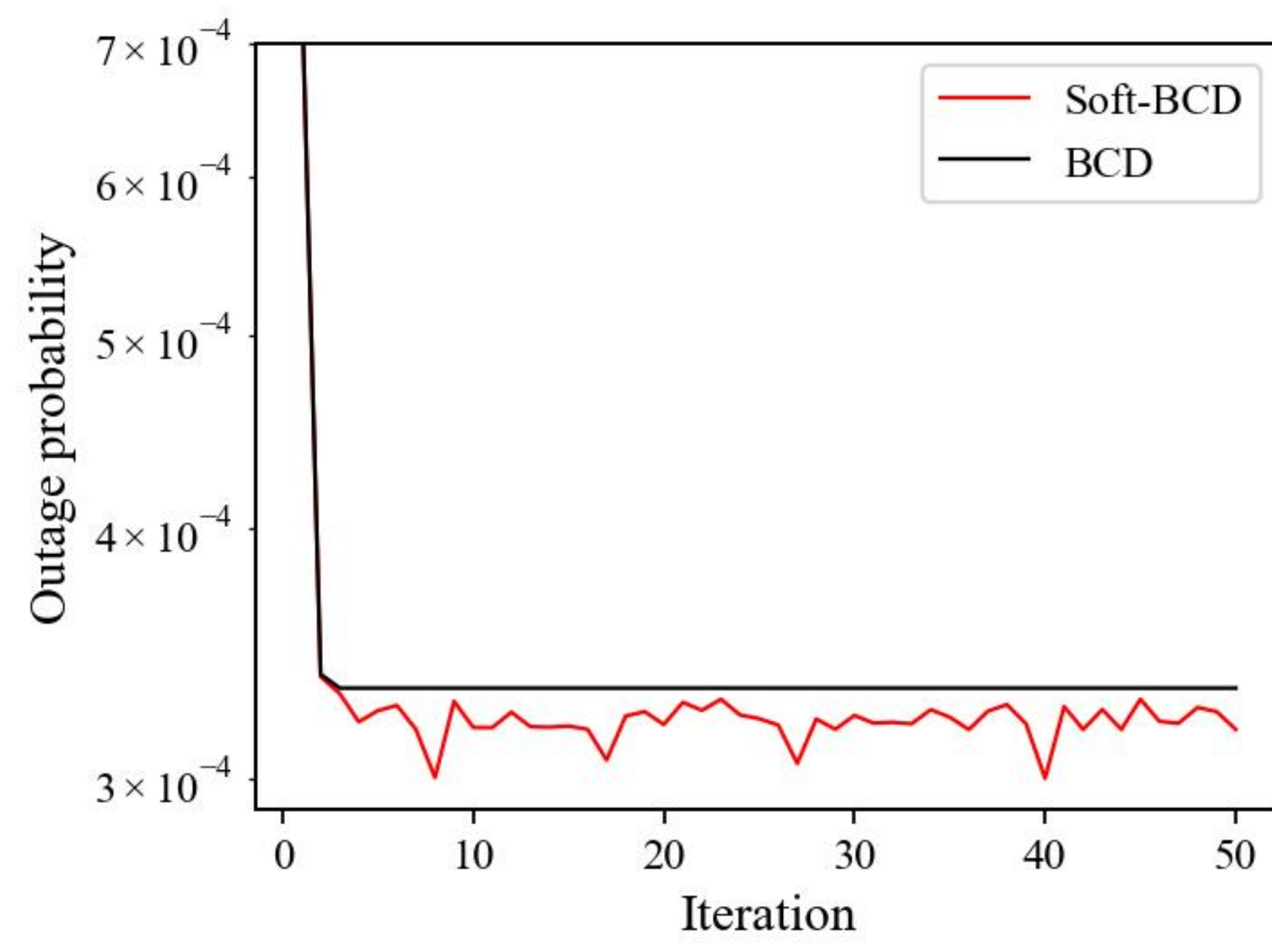




Fig. 4: Strategy comparison versus relay transmit power.

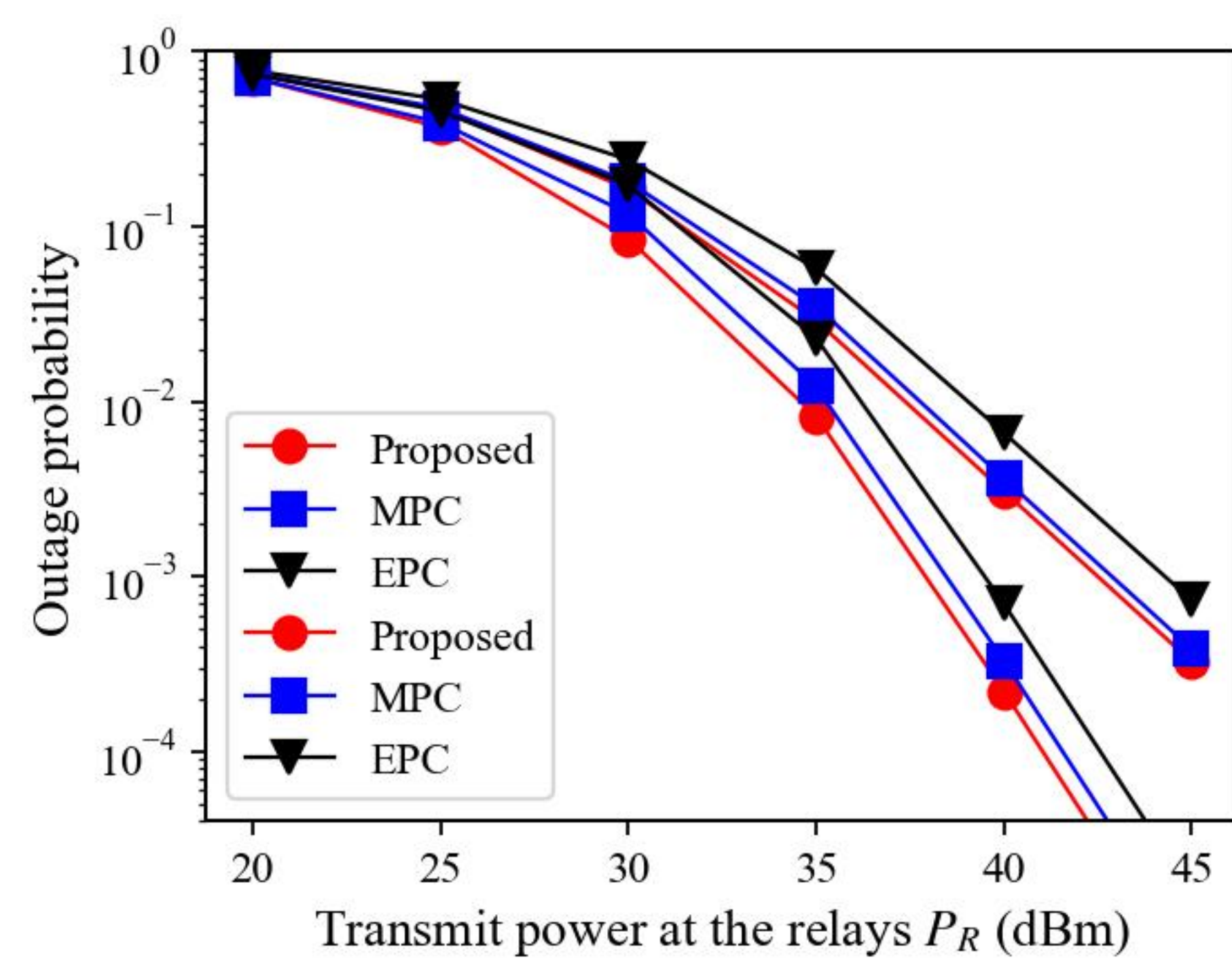




Fig. 5: Strategy comparison versus BS transmit power.

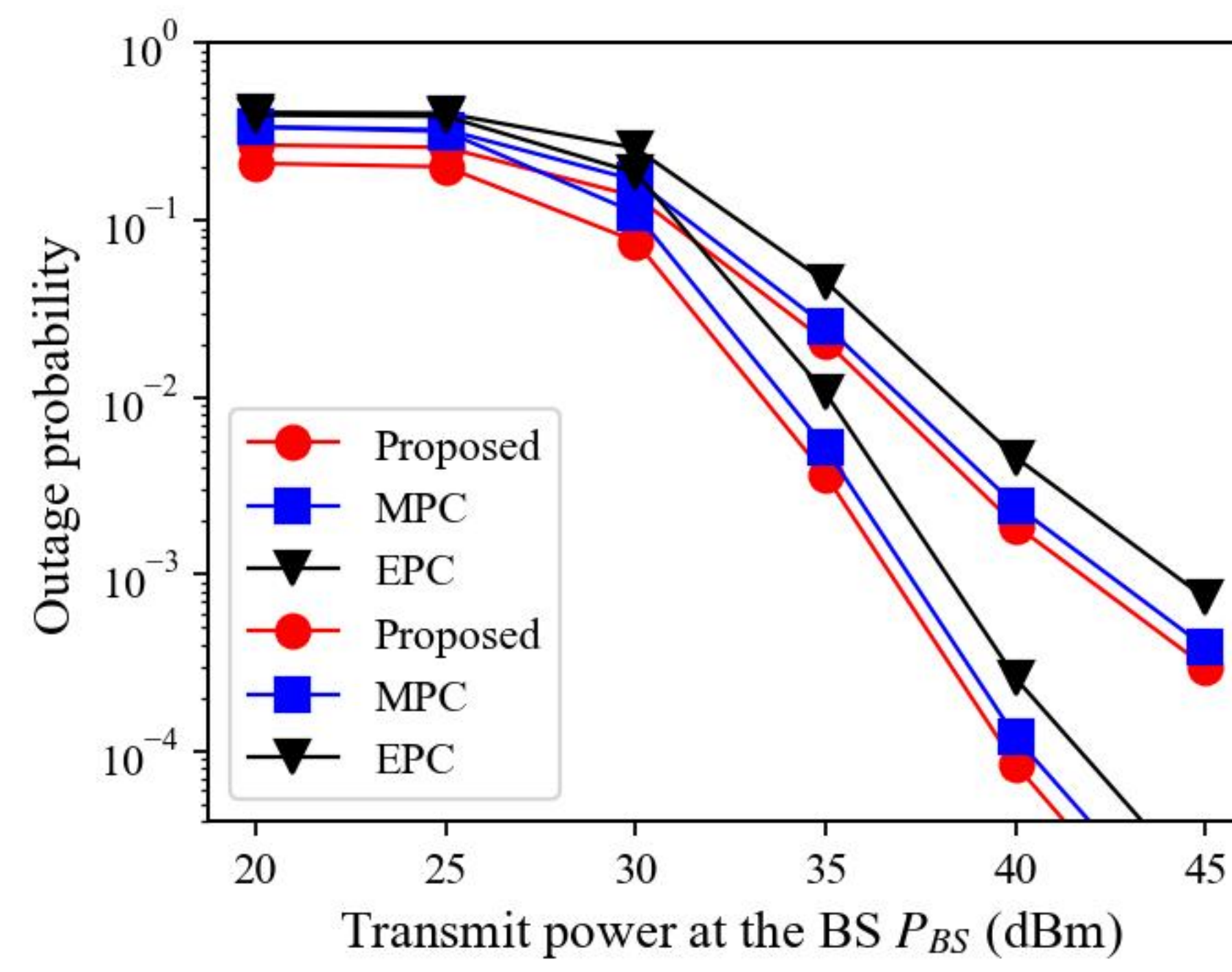




Fig. 6: Strategy comparison versus relay cache size  $C_1$ .

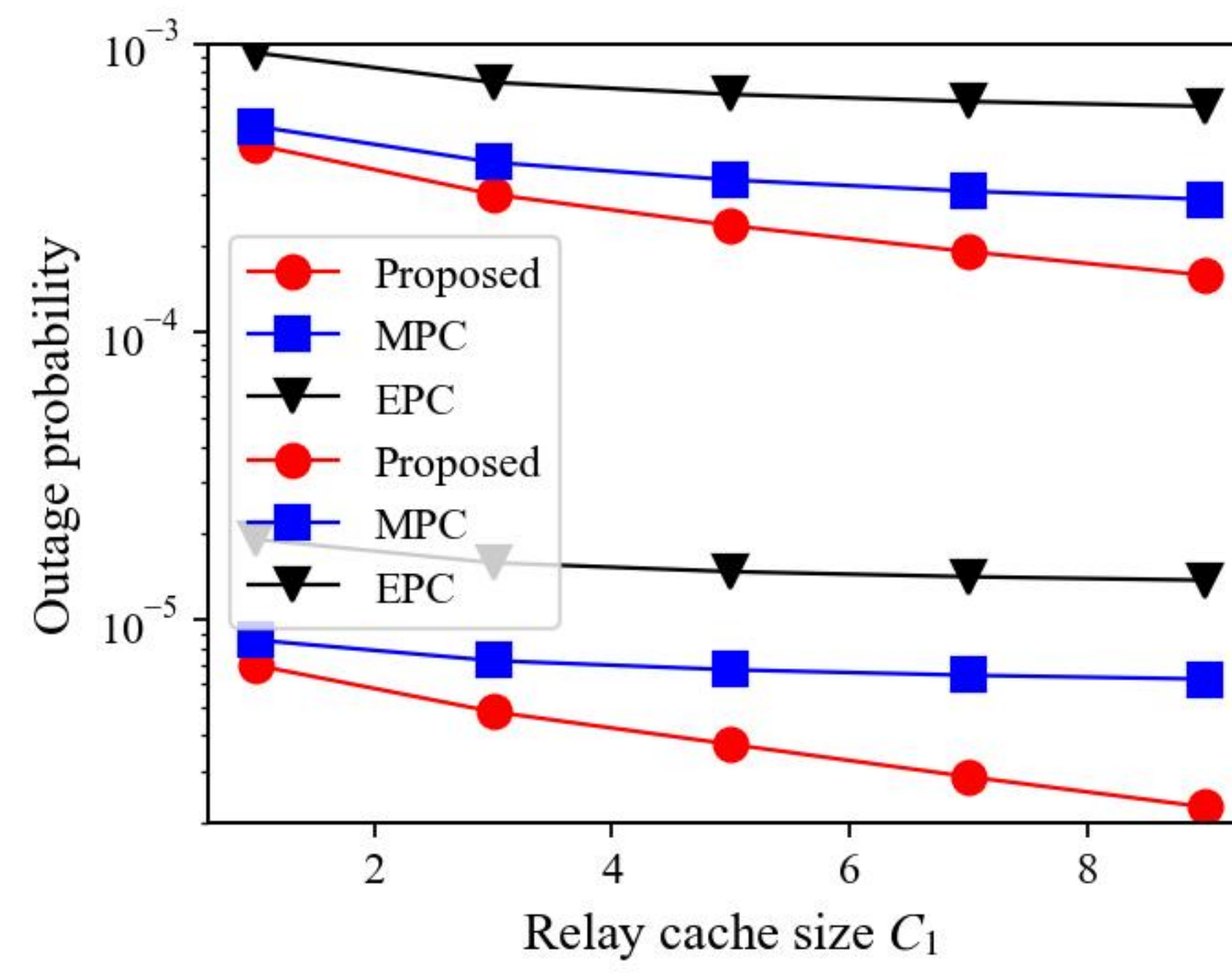




Fig. 7: Strategy comparison versus BS cache size  $C_2$ .

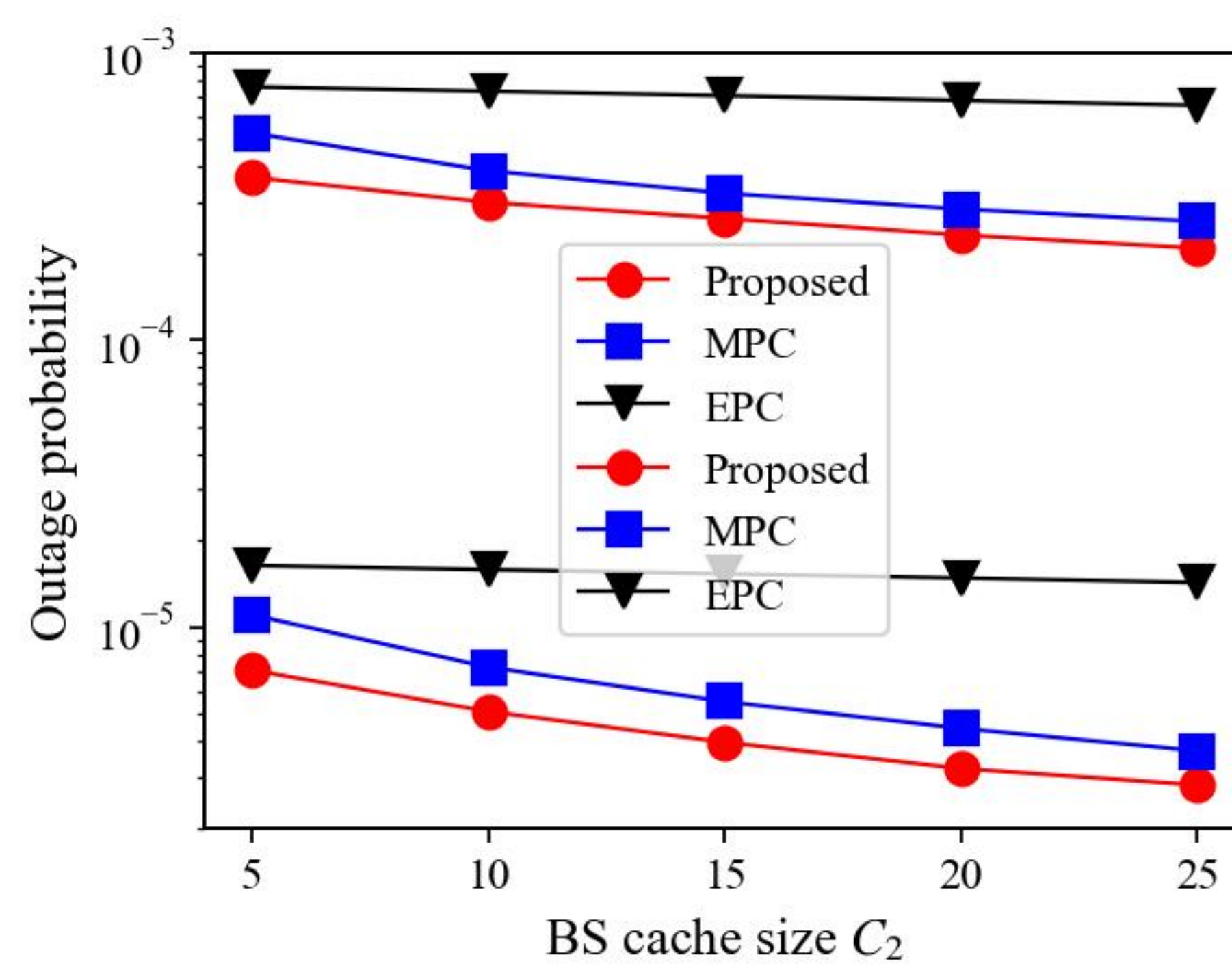




Fig. 8: Strategy comparison versus MZipf skewness eta.

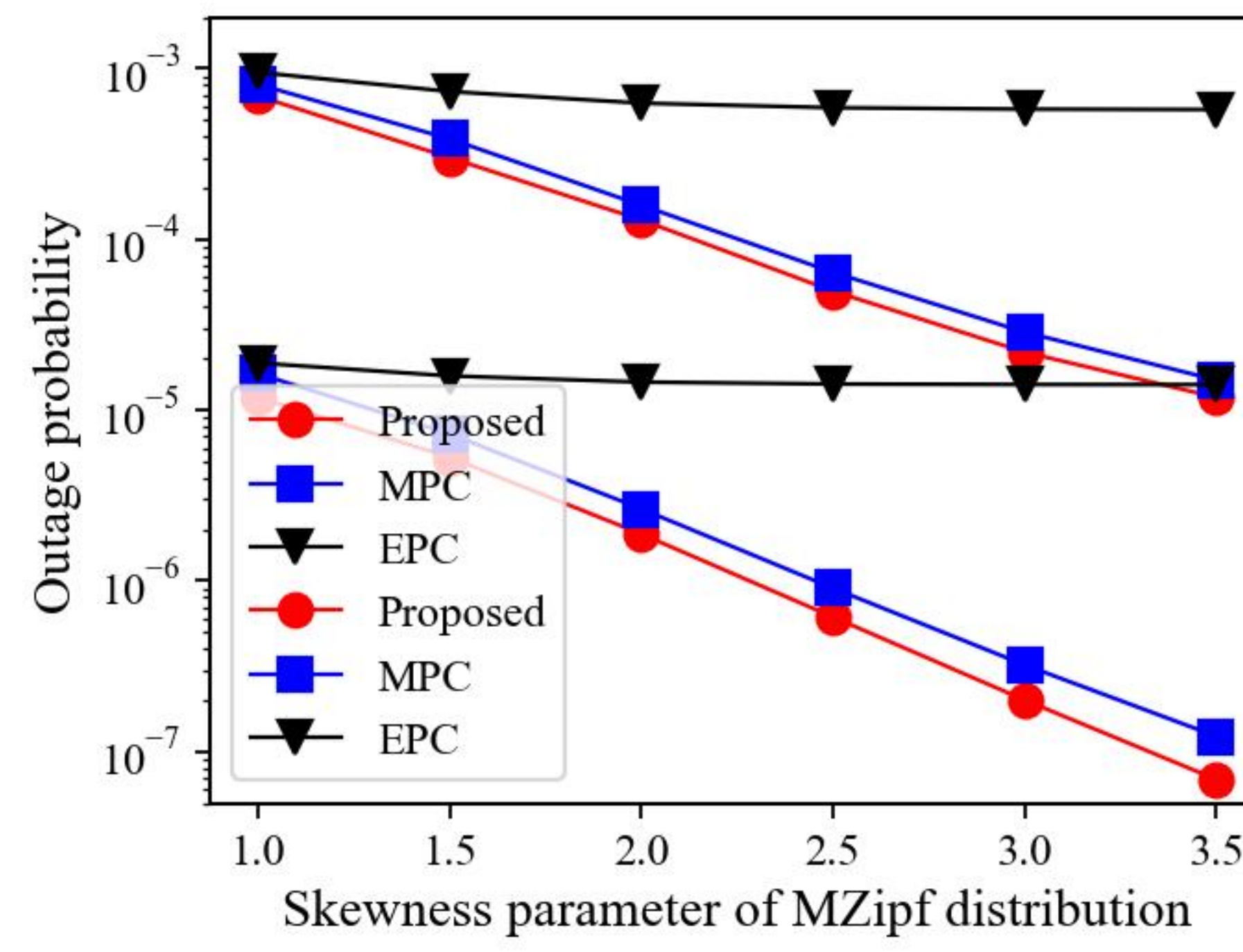




Fig. 9: Strategy comparison versus MZipf plateau tau.

